

Deepfake Detective & AI Safety · Full Curriculum

Single session (2.5 hrs) · 12-16 yrs · Intermediate

Full facilitator curriculum **PREMIUM**

Overview

Critical thinking meets cutting-edge AI: spot fakes, understand misinformation, and write a personal AI safety code.

DETAIL

Track	AI Skills for Kids
Format	Single session (2.5 hrs)
Ages	12-16 yrs
Level	Intermediate
Price	from KES 5,000
Standalone	Yes
Prerequisites	None — standalone

Course Overview

Deepfake Detective & AI Safety is a single, intensive 2.5-hour workshop that turns students into critical-thinking investigators for the AI age. Over four connected mini-sessions, students learn to spot manipulated images, video and audio; trace how a fake story travels from a single post to a viral news event; build their own working "Fake or Real?" image classifier using Google's Teachable Machine; and finish by writing a personal code of conduct for using AI responsibly, which they then test in a live classroom debate.

The arc of the day moves from *detection* (spotting fakes with the naked eye) to *systems thinking* (understanding why and how misinformation spreads) to *creation* (training their own AI model) to *reflection and voice* (writing values and defending them out loud). Each part builds the vocabulary and confidence needed for the

next — students can't debate AI ethics convincingly until they've felt how convincing a deepfake can be, and they can't appreciate classifier accuracy until they've traced how fast a false claim can spread unchecked.

By the end of the session, every student takes home two physical/digital artefacts: a hand-illustrated **AI Safety Code poster** stating five personal rules for using and sharing AI content responsibly, and a working, testable **Fake-or-Real image classifier** (built in Teachable Machine, with a shareable link and screenshot log of test results) that they can show family and friends.

Learning Outcomes

- Identify at least five visual or audio artefacts commonly found in AI-generated/deepfake media (e.g. eye/blink irregularities, hand and ear distortion, lighting mismatches, unnatural audio pacing).
- Apply a structured "detective checklist" and basic reverse-image-search technique to judge whether a piece of media is likely real, fake, or unverifiable.
- Explain, using a real or simulated case, how a piece of misinformation spreads across platforms (origin → share → remix → pickup → correction) and name at least two factors that accelerate viral spread.
- Collect a labelled image dataset, train an image classification model in Teachable Machine, and evaluate its accuracy on new test images.
- Describe at least two limitations or biases of AI classifiers (e.g. dataset bias, overfitting to specific lighting/backgrounds).
- Draft a personal AI Safety Code of five actionable rules and justify each rule with a reason.
- Construct and deliver a spoken argument for or against a real ethical debate topic, and respond respectfully to a counter-argument.

Materials Master List

ITEM	SPEC	QTY PER STUDENT/ GROUP	NOTES
Laptop or Chromebook	Webcam, Chrome browser, stable internet/Wi-Fi, min. 4GB RAM	1 per student (or 1 per pair for Session 3)	Pre-test Wi-Fi and Chrome access before the session;

Teachable Machine runs entirely in-browser

Facilitator Media Pack (USB or shared drive)	6–8 curated real/fake images, 1 real video clip, 1 deepfake video clip, 1 real audio clip, 1 AI voice-clone audio clip, all saved locally as files	1 pack for whole class, projected/ shared	Prepare in advance — see Session 1 build detail; do not rely on live internet to source these on the day
Facilitator Dataset Pack	Folder with 40 real human face photos + 40 AI-generated face photos (e.g. sourced from a public-domain face set and an AI image generator, saved as JPG/PNG in advance)	1 shared folder, copied to each laptop/pair	Used for Teachable Machine training in Session 3; keep filenames simple (real_01.jpg, ai_01.jpg)
Detective Case File worksheet (printed)	A4, double-sided, one row per media item with checklist boxes	1 per student	Template detailed in Session 1
Misinformation Trace Cards (printed, cut)	A6 cardstock, 8 cards per set (Origin post, First share, Meme remix, Influencer post, News pickup, Fact-check article, Correction, Platform algorithm boost)	1 set per group of 3–4	Laminate for reuse across cohorts
Large flipchart paper (A2)	Plain or grid	1 per group	For the misinformation spread map
Markers / sketch pens	Assorted colours, fine + broad tip	1 set (6–8 colours) per group	Nerokas or Pixel Electronics stock these; any stationery supplier works
Sticky notes	76x76mm, assorted colours	1 pad per group	Used for annotating trace maps and debate arguments
AI Safety Code poster template (printed)	A3, pre-printed with title banner and 5 numbered rule boxes plus illustration space	1 per student	Take-home artefact — see Session 4
Coloured pencils / fine markers	Assorted	1 set per student	For decorating the poster
Debate role cards (printed)	A5, "FOR" / "AGAINST" / "Moderator note" cards	1 per student	See Session 4
Argument builder worksheet (printed)	A4, claim/evidence/reasoning boxes	1 per student	Debate prep scaffold
Stopwatch or timer (phone/laptop)	Any	1 for facilitator	Keep debate rounds strictly timed

Whiteboard/ flipchart + markers	Standard classroom whiteboard	1 for facilitator	For live demos and vocabulary
Projector or large screen	HDMI-connected, min 720p	1 for room	To display media pack and Teachable Machine demo live

Session Plans

Part 1: Spot the fakes

Objectives

- Identify common visual and audio artefacts in AI-generated/deepfake media.
- Apply a structured detective checklist to rate media as Real, Fake, or Unsure with written evidence.
- Practice a basic lateral-reading/reverse-image-search move to verify a claim.

Timing (35 minutes)

- 0–5 min — Hook: side-by-side "spot the difference" reveal
- 5–10 min — Mini-lecture: what is a deepfake, how are they made (GANs/diffusion, voice cloning)
- 10–15 min — Detective checklist taught with live demo on one example
- 15–30 min — Small-group case file activity (6 media items)
- 30–35 min — Reveal, debrief, and reverse-image-search demo

Step-by-step

1. Project two portrait photos side by side (one real, one AI-generated, from the Media Pack). Ask students to vote by show of hands on which is fake. Do not reveal the answer yet — tell them they'll find out by the end of this part.
2. Explain in plain language: a deepfake is media (image, video or audio) generated or altered by AI to look/sound real. Briefly mention the two main techniques without going deep into math: GANs (two AI systems compete — one generates, one detects) and diffusion models (an AI "denoises" random static into an image based on a text prompt). For audio, mention voice cloning — an AI trained on a few minutes of someone's voice can generate new sentences in that voice.

3. Introduce the "Detective Checklist" on the whiteboard. Write these categories: Eyes & blinking, Hands & fingers, Ears & hair strands, Teeth, Lighting & shadows, Background warping, Skin texture (too smooth/waxy), Audio-lip sync, Voice pacing/pitch, Context clues (does the claim make sense? is it dated/sourced?).
4. Demo: pick one video clip from the Media Pack. Pause it and think aloud through the checklist — "Look at her earrings, they change shape between frames... the blinking pattern looks too regular... this is a red flag." Model uncertainty too — say "I'm not 100% sure, so I'd rate this Unsure and go verify."
5. Split students into groups of 3–4. Hand out the Detective Case File worksheet (6 rows, one per media item in the Media Pack — 2 images, 2 video clips, 2 audio clips, mixed real/fake). Groups rotate through stations (or watch clips projected in sequence) and fill in: Rating (Real/Fake/Unsure), Evidence (which checklist clue), Confidence (1–5).
6. Reveal the correct answers one by one. For each, ask one group to share their evidence. Tally how many groups got each one right — this usually shows video/audio are harder to judge than still images.
7. Close with a 3-minute live demo of a reverse-image-search technique (right-click → search image, or a lens-style image search tool) on one of the still images, showing how to find the image's original source and context. Explain "lateral reading" — instead of staring harder at the image, leave it and check what other trustworthy sources say about it.

Build detail

This part has no live "build" — its output is a completed investigation, so preparation happens before the session. The facilitator must assemble a **Media Pack** at least a day in advance because deepfake-detection websites are unreliable to load live and some AI-image sites can be geo-restricted or slow:

- **2 real photographic portraits** — any royalty-free stock photo set.
- **2 AI-generated portraits** — generate ahead of time using an AI image tool (e.g. Bing Image Creator, Craiyon, or Midjourney) with prompts like: "photorealistic portrait of a smiling young woman, studio lighting, neutral grey background" and "photorealistic portrait of an older man wearing glasses, outdoor daylight." Save both as JPG files.
- **1 real, unedited short video clip** (10–20 seconds) — any authentic news or interview clip saved locally.

- **1 deepfake or face-swapped video clip** — use a well-known, publicly archived deepfake awareness/education example (many fact-checking organisations publish these openly for teaching purposes); download and store locally before the session.
- **1 real audio clip** — a short recording of a real person speaking.
- **1 AI voice-clone audio clip** — generate ahead of time using a text-to-speech/voice-cloning tool (e.g. ElevenLabs free tier) with the sample text: "Hey, it's me — can you send me the homework answers before class?" Save as MP3.

Store everything in one folder titled *MediaPack*, numbered 1–8, with an answer key kept separately (facilitator eyes only) noting which are Real/Fake. A good result for this activity is a completed worksheet where each group has written specific, checklist-based evidence (not just a guess) for at least 5 of the 6 items, and can articulate in the debrief why video and audio are harder to judge than still images.

Checks for understanding

- "Name two things you'd look at on a person's face to judge if a video is a deepfake."
- "Why might a still image be easier or harder to fake-detect than a video?"
- "What does 'lateral reading' mean, and why is it useful even when you can't spot visual clues?"

Common mistakes & fixes

MISTAKE	FIX
Students rate everything "Fake" because they're primed to distrust	Remind them the checklist includes real items too; ask "what specific evidence supports that, not just suspicion?"
Groups rush through without discussing evidence	Require at least one written clue per row before moving to the next item; circulate and prompt with "what exactly made you say that?"
Confusing "edited/filtered" photos with "AI-generated" photos	Clarify the distinction on the whiteboard: a filter changes an existing real photo, an AI-generated image was never a real photo at all

Part 2: Trace misinformation

Objectives

- Explain the difference between misinformation, disinformation and malinformation.

- Sequence and map the life cycle of a fake story from origin to viral spread.
- Identify at least two factors (emotional content, algorithm boosting, influencer reach) that accelerate spread.

Timing (30 minutes)

- 0–5 min — Mini-lecture: three types of false information
- 5–12 min — Present the simulated case study
- 12–25 min — Group activity: sequence and map the trace cards
- 25–30 min — Group share-outs and discussion

Step-by-step

1. On the whiteboard, write three terms: Misinformation (false info shared without knowing it's false), Disinformation (false info shared deliberately to deceive), Malinformation (true info shared out of context to cause harm). Give one quick example of each verbally.
2. Introduce a facilitator-created, fictional-but-realistic case study to avoid teaching real defamatory content: "*Breaking: Strange lights spotted over Uhuru Park last night*". Explain this is invented for the exercise, but the pattern of spread is exactly how real misinformation travels.
3. Hand each group of 3–4 the shuffled set of 8 Misinformation Trace Cards: *Origin post* (an anonymous account posts a blurry photo with the claim), *First share* (a friend forwards it on WhatsApp), *Meme remix* (someone edits it into a joke meme), *Influencer post* (a large account reposts it as "breaking news" without checking), *Platform algorithm boost* (the post gets recommended widely because of high engagement), *News pickup* (a small blog writes an article based on the post), *Fact-check article* (a fact-checking site investigates and finds it's a drone light show), *Correction/retraction* (the original poster deletes it, but slowly).
4. Groups must arrange the 8 cards in the order they think this story most likely spread, then glue/tape them onto the A2 flipchart paper and draw arrows between them labelled with the platform (WhatsApp, Twitter/X, TikTok, news site) and an estimate of how many people saw it at that stage (they can guess reasonable numbers, e.g. 5 → 200 → 5,000 → 50,000).
5. Circulate and ask probing questions: "Why do you think the meme spread faster than the correction?" "What made the influencer share it without checking?"

6. Bring the class together. Have 2 groups present their maps. Draw out the key insight on the whiteboard: corrections almost always travel slower and reach fewer people than the original false claim — this is sometimes called the "correction gap."

Build detail

There is no digital tool build in this part — the "build" is the physical trace map. Facilitator preparation: print and laminate one set of 8 Trace Cards per group in advance (reusable across cohorts). Keep the case study fictional and low-stakes (lights, aliens, a viral prank) rather than a real political or health claim, so the lesson stays about the mechanism of spread, not about taking sides on a real controversial event. A good result is a flipchart map with all 8 stages placed in a defensible order, arrows labelled with platform names, and at least one annotation explaining WHY a stage spread quickly (e.g. "shocking," "shared by someone trusted," "algorithm pushed it to more feeds").

Checks for understanding

- "What's the difference between misinformation and disinformation?"
- "In our trace map, which stage caused the story to reach the most people, and why?"
- "Why does the correction usually reach fewer people than the original claim?"

Common mistakes & fixes

MISTAKE	FIX
Groups place the Fact-check card first (assuming corrections happen quickly)	Ask "would anyone fact-check something before it's even popular?" to guide them to place it later
Confusing misinformation with disinformation during discussion	Point back to the whiteboard definitions and ask "did the original poster know it was false?"
Groups don't label arrows with platforms/ numbers	Give a 1-minute warning at 20 minutes: "if your map has no platform names or numbers yet, add them now before we share"

Part 3: Build a classifier

Objectives

- Collect and label a small image dataset for a two-class classification problem.

- Train and test an image classification model using Teachable Machine.
- Evaluate the model's accuracy and identify at least one limitation of the trained model.

Timing (55 minutes)

- 0–4 min — Intro: today you're building the "Fake or Real?" quiz brain
- 4–9 min — Live facilitator demo: creating a Teachable Machine project
- 9–13 min — Defining the two classes together
- 13–28 min — Dataset collection and upload (using Dataset Pack + own webcam photos)
- 28–33 min — Training the model
- 33–41 min — Testing and iterating
- 41–50 min — Pair up, run each other's quiz, guess-then-reveal
- 50–55 min — Wrap-up discussion: accuracy and bias

Step-by-step

1. Explain the goal: each pair will train an AI model that guesses whether a face photo is Real or AI-generated, then challenge a classmate to beat the AI at the same guessing game.
2. On the projector, open Teachable Machine (teachablemachine.withgoogle.com) and click **Get Started** → **Image Project** → **Standard image model**. Show students the interface: Class boxes on the left, Training button in the middle, Preview panel on the right.
3. As a class, agree on two class names: **Class 1: "Real"** and **Class 2: "AI-Generated."** Type these into the class name boxes on the projected demo so everyone copies the same naming (consistent naming avoids confusion later when comparing results).
4. Students work in pairs on one laptop. Each pair opens their own Teachable Machine project in a new browser tab and names their two classes exactly the same way.
5. Dataset collection: instruct each pair to click **Upload** under "Real" and select at least 15 images from the shared Dataset Pack's "real" folder, and click **Upload** under "AI-Generated" and select at least 15 images from the "ai" folder. Encourage pairs who finish early to also use the webcam button to add 3–5 live photos of themselves to the "Real" class for a personal touch.

6. Click **Train Model**. While it trains (usually under a minute for this dataset size), explain simply: "The model is looking at patterns of pixels in each class over and over — this is called an epoch — until it gets good at telling them apart."
7. Once training finishes, use the live **Preview** panel: hold up a printed test photo (not used in training) to the webcam, or upload a fresh test image using the "File" preview option. Read the confidence percentages shown (e.g. "Real: 82%, AI-Generated: 18%").
8. Have pairs test at least 5 new images they haven't trained on (facilitator hands out a small "Test Set" — 5 held-back images not in the Dataset Pack's training folders) and record each result on a simple tally sheet: correct/incorrect.
9. If accuracy is poor, guide iteration: add more training images to the weaker class, remove any mislabeled or blurry images, and retrain. This mirrors real machine-learning workflow.
10. Pairs swap laptops with a neighbouring pair. Each pair shows 3 unseen photos on their screen; the neighbouring pair must guess Real/AI first, then the model reveals its own guess. Discuss: did the AI beat the humans?
11. Class wrap-up discussion: ask why the model might get confused (same-race/ gender clustering in Dataset Pack, lighting similarities, image resolution) — this introduces the idea of **dataset bias**.

Build detail

Tool: Teachable Machine (teachablemachine.withgoogle.com), a free, no-install, browser-based tool from Google. No account or sign-in is required to build and test a model; a free Google account is only needed if students want to save/export a shareable link.

Setup before the session: confirm every laptop can load the site over the venue Wi-Fi (test at least one hour before class), and pre-load the Dataset Pack folder (40 real + 40 AI faces, plus a separate 5–10 image "Test Set" not shown during training) onto each laptop's desktop via USB, since uploading over slow Wi-Fi is the single biggest time sink.

Exact workflow: Get Started → Image Project → Standard image model → name Class 1 "Real" → name Class 2 "AI-Generated" → Upload 15–20 images per class from local folders (webcam capture optional add-on) → Train Model → Preview with webcam or file upload → read confidence bars → iterate by adding/removing

images and retraining → (optional) Export Model → Upload (shareable link) to generate a public link students can save and show family at home.

What a good result looks like: a trained model that correctly classifies at least 4 out of 5 held-back test images, and a pair of students who can explain in their own words one reason the model might get one wrong (e.g. "the lighting in this photo looks similar to our AI training photos").

Testing/iteration: the built-in tally sheet (correct vs incorrect out of 5 test images) is the iteration metric — if accuracy is below 3/5, students add 5–10 more images to the weaker class and retrain before moving on.

Take-home export (optional stretch): for pairs who want to embed their trained model into a shareable webpage quiz, show this complete working example, which loads any Teachable Machine model by its public link and displays live predictions:

```
<!DOCTYPE html>
<html>
<head>
  <title>Fake or Real? Quiz</title>
  <script src="https://cdn.jsdelivr.net/npm/@tensorflow/tfjs@1.3.1/dist/tf.min.js"></script>
  <script src="https://cdn.jsdelivr.net/npm/@teachablemachine/image@0.8/dist/teachablemachine-
image.min.js"></script>
</head>
<body>
  <h1>Fake or Real? Quiz</h1>
  <button type="button" onclick="init()">Start Webcam Test</button>
  <div id="webcam-container"></div>
  <div id="label-container"></div>

  <script type="text/javascript">
    // Replace this URL with the link from your own Teachable Machine model
    const URL = "https://teachablemachine.withgoogle.com/models/YOUR_MODEL_ID/";

    let model, webcam, labelContainer, maxPredictions;

    async function init() {
      const modelURL = URL + "model.json";
      const metadataURL = URL + "metadata.json";

      model = await tmImage.load(modelURL, metadataURL);
      maxPredictions = model.getTotalClasses();

      const flip = true;
      webcam = new tmImage.Webcam(300, 300, flip);
      await webcam.setup();
      await webcam.play();
      window.requestAnimationFrame(loop);

      document.getElementById("webcam-container").appendChild(webcam.canvas);
      labelContainer = document.getElementById("label-container");
      for (let i = 0; i < maxPredictions; i++) {
```

```

        labelContainer.appendChild(document.createElement("div"));
    }
}

async function loop() {
    webcam.update();
    await predict();
    window.requestAnimationFrame(loop);
}

async function predict() {
    const prediction = await model.predict(webcam.canvas);
    for (let i = 0; i < maxPredictions; i++) {
        const classPrediction =
            prediction[i].className + ": " + prediction[i].probability.toFixed(2);
        labelContainer.childNodes[i].innerHTML = classPrediction;
    }
}
</script>
</body>
</html>

```

Facilitator note: students only need to replace `YOUR_MODEL_ID` with their own model's link (found via Teachable Machine's Export Model → Upload my model → copy shareable link). This file can be saved as `quiz.html` and opened directly in any browser — no server required.

Checks for understanding

- "What does it mean when the model says 'Real: 60%, AI-Generated: 40%'?"
- "Why did we test on images the model had never seen before, instead of images we trained it on?"
- "Name one reason your model might get an answer wrong."

Common mistakes & fixes

MISTAKE	FIX
Uploading too few images per class (under 10), causing wildly inaccurate results	Set a minimum of 15 images per class before allowing training; circulate to check counts
Testing only on images already used in training, inflating apparent accuracy	Always use the separate, held-back Test Set for evaluation; explicitly ban testing on training images
Students think the model "understands" images the way humans do	Clarify it's matching pixel patterns statistically, not reasoning about content — this connects to the bias discussion

Part 4: AI Safety Code & debate

Objectives

- Draft a personal AI Safety Code of five clear, actionable rules.
- Construct a spoken argument with claim, evidence and reasoning for a real debate motion.
- Practice respectful listening and rebuttal in a structured debate format.

Timing (30 minutes)

- 0–3 min — Recap: from detecting fakes to deciding what YOU will do
- 3–13 min — Individual poster drafting: My AI Safety Code
- 13–16 min — Pair-share and borrow one idea
- 16–21 min — Debate setup: sides, roles, prep time
- 21–29 min — Structured debate (openings, rebuttals, floor)
- 29–30 min — Class vote and closing reflection

Step-by-step

1. Bring the room back together. Say: "Today you learned to spot fakes, trace their spread, and even build an AI that judges real vs fake. Now: what rules will YOU personally follow when you use AI or see AI content online?"
2. Hand out the AI Safety Code poster template (A3, 5 numbered rule boxes). Give sentence starters on the whiteboard to scaffold thinking: "I will always check ___ before I share something," "I will never use AI to ___," "If I'm not sure if something is real, I will ___," "When I use AI to create something, I will ___," "If a friend shares a fake, I will ___."
3. Students work individually for 10 minutes, writing one rule per box in their own words and adding a small doodle/icon per rule. Circulate and prompt quieter students with "what would you tell your younger sibling to watch out for?"
4. Pair students up for 3 minutes to read each other's posters aloud and each add one new rule (or refine one) inspired by their partner, in a 6th "bonus rule" box if space allows, or by annotating an existing rule.
5. Announce the debate motion, written large on the whiteboard: "**Should AI-generated art be allowed in school (for projects, posters, assignments)?**" Split the class into two halves — FOR and AGAINST — by

counting off or by card draw (this avoids students always picking their pre-existing opinion, which produces livelier debate).

6. Hand out the Argument Builder worksheet. Each side has 5 minutes to write at least 2 arguments using Claim → Evidence → Reasoning format, drawing on what they learned earlier in the day (e.g. "Claim: AI art can be a tool like a paintbrush. Evidence: photography was once seen as 'not real art' too. Reasoning: new tools change art, they don't destroy it" versus "Claim: AI art can be plagiarism. Evidence: AI models are trained on other artists' work without consent. Reasoning: using it in school normalises taking credit for others' effort").
7. Run the structured debate as moderator, timing strictly: FOR opening statement (2 min) → AGAINST opening statement (2 min) → FOR rebuttal (2 min) → AGAINST rebuttal (2 min) → open floor for 2–3 additional voices per side (3 min) → each side's closing 1-line statement (1 min total).
8. Call a class vote by show of hands ("who was more convinced by the end than at the start?") and close with a 1-minute reflection: "Notice — this was a real debate with no easy answer. That's exactly why your AI Safety Code matters: it's your own compass for situations with no single right answer."

Build detail

The "build" here is the AI Safety Code poster itself — treat it with the same rigour as a technical build. Provide the pre-printed A3 template with a title banner ("MY AI SAFETY CODE") and 5 numbered boxes so every student produces a consistent, presentable take-home artefact. A good result is a poster where every one of the 5 rules is specific and actionable (not vague like "be careful" but concrete like "I will reverse-image-search any shocking photo before I forward it") and at least one rule directly references something learned earlier in the session (detective checklist, spread/correction gap, or classifier bias). For the debate, a good result is every student speaking at least once (opening, rebuttal, or floor comment) and using the Claim → Evidence → Reasoning structure at least once.

Checks for understanding

- "Read me one rule from your AI Safety Code and explain why you chose it."
- "What's one argument from the OTHER side of the debate that you found convincing, even if you disagreed overall?"
- "How did what you learned earlier today (checklist, spread map, or classifier) show up in your rules or arguments?"

Common mistakes & fixes

MISTAKE	FIX
Rules are too vague ("Don't be fooled by fakes")	Push for a specific action verb and trigger: "When I see ___, I will ___"
Debate turns into shouting or personal attacks	Stop immediately, restate the rule "attack the argument, not the person," and restart the clock for that round
One or two students dominate the floor discussion	Use a "one voice per side per round" rule and call on quieter students by name during the open floor

Assessment & Showcase

Assessment across the session is formative and artefact-based rather than test-based — the facilitator collects evidence of learning through four checkpoints as the day progresses:

- **Detective Case File worksheet** (Part 1) — checked for at least 5 of 6 rows completed with specific, checklist-based evidence, not just guesses.
- **Misinformation trace map** (Part 2) — checked for a defensible sequence, labelled platforms/arrows, and at least one written explanation of why a stage spread quickly.
- **Trained classifier + test tally sheet** (Part 3) — checked for a trained model, a completed 5-image test tally, and a verbal or written note on one limitation/bias observed.
- **AI Safety Code poster + debate participation** (Part 4) — checked for 5 specific, actionable rules and at least one spoken contribution during the debate.

Run a 10–15 minute closing showcase (can extend the session slightly or fold into Part 4's final minutes if running tight): set up a "gallery walk" where posters are displayed on desks/walls and 2–3 pairs volunteer to plug in their laptop and run a live 60-second demo of their Teachable Machine quiz on 2 fresh photos in front of the class, inviting classmates to guess before the model reveals its answer. Close with a class vote for "Most Convincing Case" (best classifier demo) and "Sharpest Rule" (best single AI Safety Code rule, read aloud by its author).

"Done" for this course means each student leaves with: a completed Detective Case File, a contribution visible on their group's trace map, a personally trained and tested Teachable Machine classifier (accessible via saved link or screenshot), a fully filled 5-rule AI Safety Code poster, and at least one recorded spoken contribution in the debate.

Extension & Home Challenges

- **Easy:** At home, find 5 new photos online (mix of real photos and AI-generated images from any free AI image generator) and run them through your saved Teachable Machine model or link. Record how many the model got right and write two sentences on which one surprised you.
- **Medium:Medium: Add a third class to your Teachable Machine model called "AI-Generated (different tool)" using images from a different AI image generator than the one in your original training set. Retrain and test — does your model still perform well, or does it confuse the two AI classes? Also, research one real, publicly documented deepfake incident (news article or fact-check site) and write a short paragraph summarising what happened and how it was eventually detected or debunked.**
- **Hard:** Pick a viral post, image or video you genuinely encountered this week (or ask a family member for one they saw). Apply the full detective checklist and a reverse-image search to investigate it, then create a 60-second video or slideshow (using your phone or any free video/slide tool) explaining your investigation and verdict, including at least one specific piece of evidence and one screenshot. As a bonus, try building a second Teachable Machine project using the Audio Project type to see if you can train a model to distinguish your own real voice recording from an AI text-to-speech clip of the same sentence.